# Project 3: Customer Loyalty

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## Scenario

Our company, **TableGram**, has strong acquisition numbers but struggles to retain customers long term. Despite frequent usage, many customers, whom we dub **"Promotional Nomads,"** exhibit loyalty challenges**,** such as switching between apps to chase discounts, discontinuing usage after promotions end, expressing dissatisfaction with delivery times or food quality. We need to understand the key **drivers** and **barriers** to loyalty and develop **loyalty-enhancing strategies**.

## Outline of Tasks

1. We designed a **survey** and created a **C-suite presentation** outlining its (simulated) insights, tied to specific business strategies.
2. Based on the simulated findings of the survey, we proposed an **A/B test** to evaluate an **intervention** aimed at improving customer loyalty, and simulated the results.
3. Following the survey and A/B test, we developed a **quantitative analysis plan** outlining the information yielded by each step and the **actionable insights** generated by this analysis.

# Survey

To capture the "Promotional Nomad," the survey needs to be punchy, mobile-optimized, and psychologically designed to uncover **why** they keep the competition's apps on their home screen.

Here is the functional survey design, structured to move a user from "Price Chaser" to "Brand Advocate."

## Survey Design

### *Section 1: The Pulse (The "Why")*

*Goal: Determine if loyalty is blocked by price, service, or lack of habit.*

**1. When you decide to order food, what is the FIRST thing you do?**

- [ ] Open TableGram immediately.
- [ ] Open whichever app has a notification/discount code.
- [ ] Check 2–3 different apps to compare total prices.
- [ ] Search Google/Maps for a specific restaurant first.

**2. Rank the following in order of importance for your "Default" app (1 = Most Important):**

- [ ] Lowest total cost (Fees + Food).
- [ ] Speed of delivery.
- [ ] Accuracy of estimated arrival time (Reliability).
- [ ] Selection of exclusive restaurants.
- [ ] Quality of customer support if something goes wrong.

### *Section 2: The Competitive Gap (The "Barrier")*

*Goal: Identify the specific "Hook" our competitors have.*

**3. Which of these statements best describes your relationship with other delivery apps?**

- [ ] I have a paid subscription (e.g., DashPass, Uber One) elsewhere.
- [ ] I only use them when they send me a "40% off" notification.
- [ ] I prefer their user interface/tracking map more than TableGram.
- [ ] I don't use other apps; I'm loyal to TableGram.

**4. What is the #1 reason you choose a competitor over us?**

- [ ] Cheaper delivery fees.
- [ ] Better rewards/points program.
- [ ] They have a specific restaurant you don't.
- [ ] Their app is faster/easier to use.

### *Section 3: The Quality Check (The "Friction")*

*Goal: Is the product "broken" or just "boring"?*

**5. In your last 3 orders with TableGram, how often did the following occur?**
*(Scale: Never | Once | Frequently)*

- Food arrived cold or messy.
- The delivery was later than the original ETA.
- Hidden fees appeared at the final checkout screen.
- Items were missing from the order.

### *Section 4: The Future (The "Strategy Builder")*

*Goal: Test which loyalty features actually move the needle for a Nomad.*

**6. Which of these "Member Perks" would make you stop checking other apps? (Select your Top 2)**

- [ ] **The "No-Fee" Pass:** A monthly subscription that pays for itself in 2 orders.
- [ ] **Priority Routing:** Your food is always the first drop-off (no "doubling up").
- [ ] **The Streak Bonus:** Get 10% back in credit for every 3 orders made in a month.
- [ ] **VIP Support:** Immediate human chat for any order issue.
- [ ] **Surprise Treats:** A free appetizer or drink from top-rated spots once a month.

### *Section 5: The "Golden Question"*

**7. If TableGram disappeared tomorrow, how disappointed would you be?**

- [ ] Very disappointed (I use it daily).
- [ ] Somewhat disappointed (I'd miss certain features).
- [ ] Not disappointed (I'd just use another app).

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## Survey Quantitative Analysis

Our survey simulated collection of **two main types** of data, and each type required a **different** approach.

### *Categorical Questions (Q1, Q3, Q4, Q6)*

These are the **multiple-choice, single-select** questions (e.g., "What is the #1 reason you choose a competitor over us?"). Each respondent picked one option, producing count data.

- **Analysis:** We started with frequency counts and percentages to see the overall distribution (this is what produces charts like "42% cite cheaper delivery fees"). To go further, we use a **chi-square test of independence** to check whether response patterns

differ across customer segments. For example: do users who hold a competitor subscription (Q3) cite different switching reasons (Q4) than users who don't?

- **How it works:** Chi-square compares the observed counts in each cell of a cross-tab table against the counts we would expect if the two variables were unrelated. If the difference is large enough ($p < .05$), we conclude the variables are associated, not independent.
- **Key assumption:** Every expected cell count must be at least 5. If any cell falls below that threshold, we switch to Fisher's Exact Test, which handles small samples without approximation.
- **What this tells us:** Whether we need **different** retention strategies for different customer segments, or whether **one strategy fits all**. If subscribers switch for fundamentally different reasons than non-subscribers, a one-size-fits-all loyalty program will underperform.

## *Ranking Question (Q2)*

**Question 2** asked users to rank five app-selection drivers (cost, speed, reliability, selection, support) from most to least important. Rank data is ordinal: we know the **order** but can't assume equal **spacing between** ranks.

- **Analysis:** We reported the mean and median rank for each attribute across all respondents. This produced the **MaxDiff-style bar chart** (Total Cost = 4.5, Reliability = 4.2, etc.). To test whether the top-ranked attributes are *significantly* more important than lower-ranked ones, rather than just appearing higher due to random variation, we used a non-parametric comparison of rank distributions.
- **What this tells us:** Whether the product team should focus on ETA accuracy vs. raw speed vs. cost reduction. The chart suggested that **reliability matters more than speed**. The statistical test tells us whether that gap is real or noise.

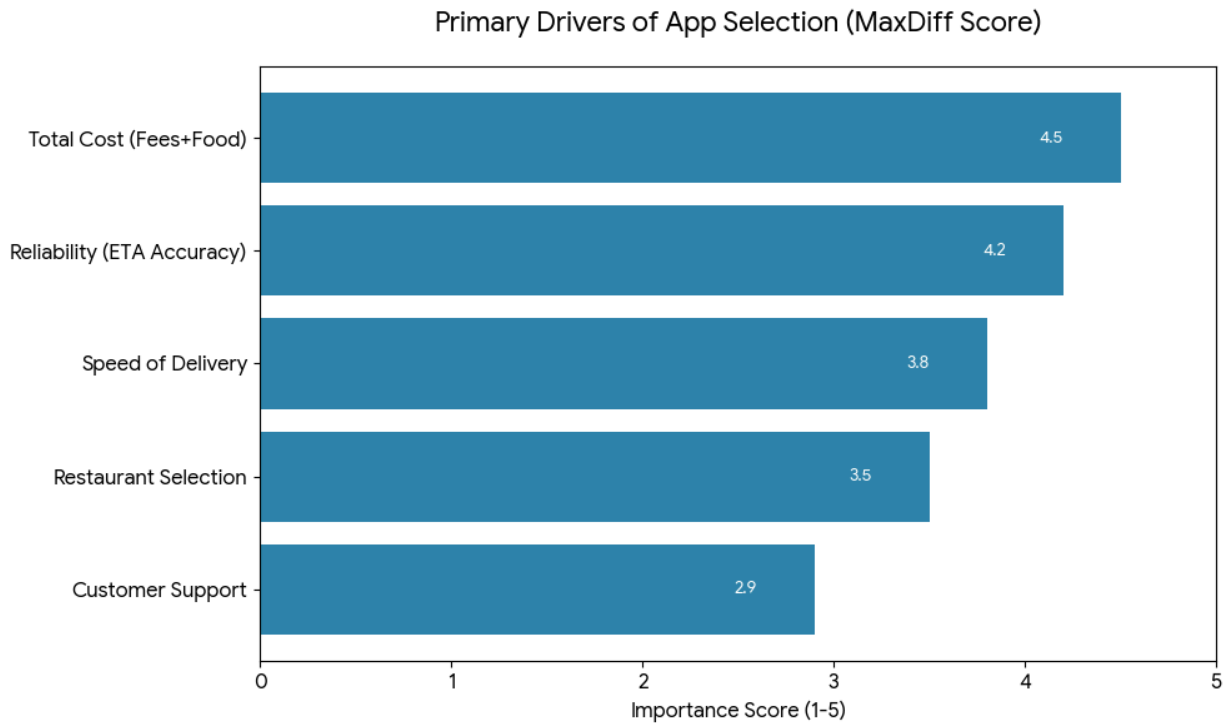
## *Friction Frequency (Q5) and Disappointment (Q7)*

**Q5** captures how often users experience **specific** problems (cold food, late delivery, hidden fees, missing items) on a Never/Once/Frequently scale. We reported the proportion of users selecting "Frequently" for each friction type and compared them. If **hidden fees** were reported as “frequent” at a significantly higher rate than other problems, that issue becomes the top-priority fix.

**Q7** (the "Golden Question") produces a **three-category distribution**: Very Disappointed / Somewhat / Not Disappointed. The key insight is the overall split. Industry benchmarks suggest that **40% "Very Disappointed"** indicates strong product-market fit. Our **10% result** quantifies the gap and gives the C-suite a concrete number to act on.

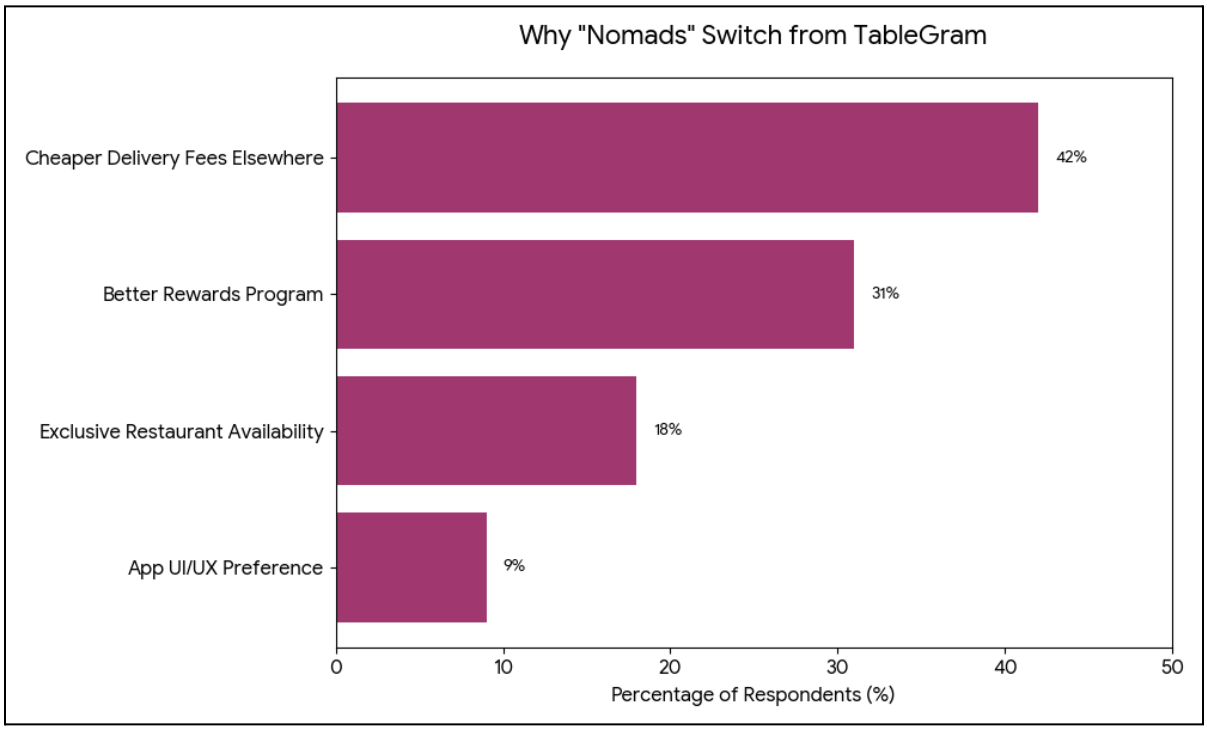
# Survey Insights

## *Chart 1: Primary Drivers of App Selection*



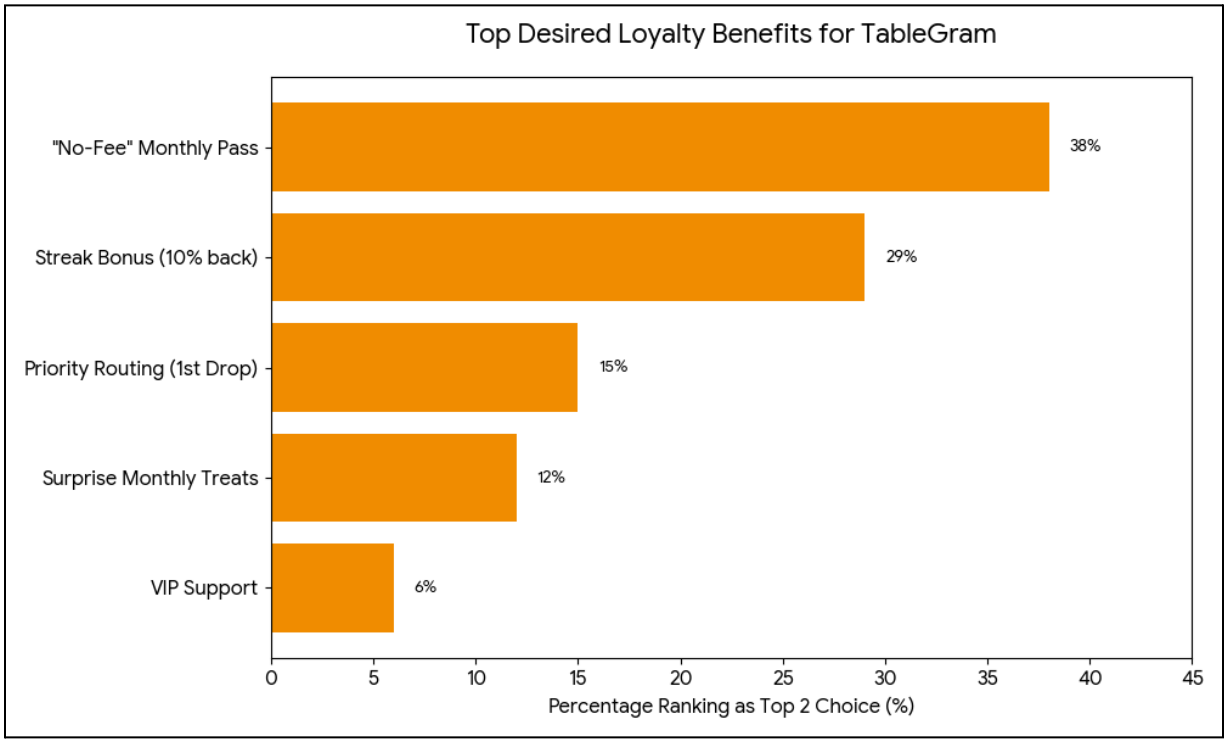

- **What it shows:** Total cost is king, but **Reliability (ETA Accuracy)** scores higher than **Speed**.
- **The C-Suite Pitch:** We don't necessarily need to be the *fastest* in the market, but we must be the most *honest*. Reliability is a driver of trust, and trust is the foundation of loyalty.

*Chart 2: Why "Nomads" Switch to Competitors*



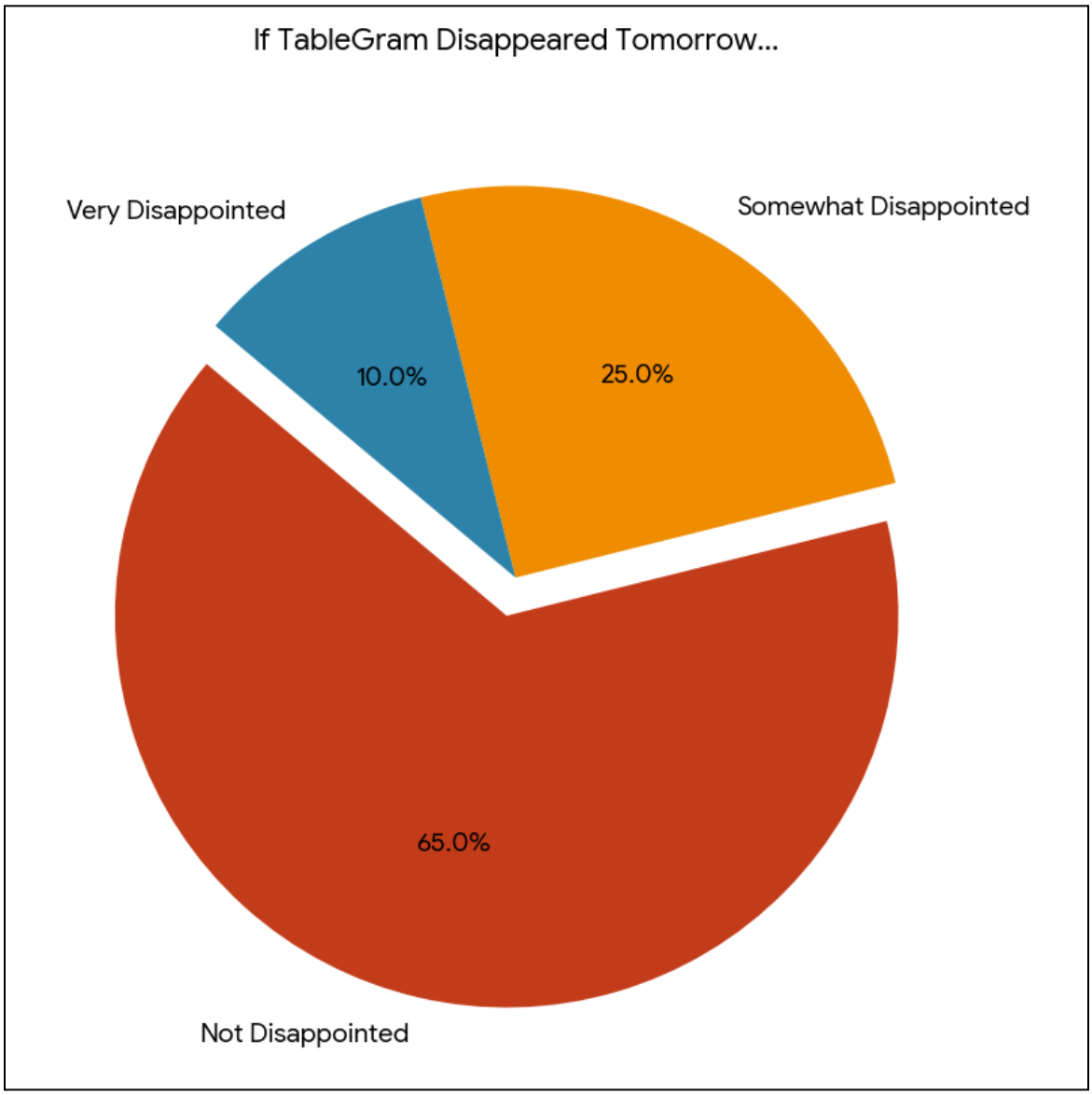

- **What it shows:** While "Cheaper Delivery Fees" is the #1 driver for switching (42%), "Better Rewards Programs" is a close second (31%).
- **The C-Suite Pitch:** We can't always win a price war on delivery fees, but we can close the 31% gap by building a **rewards ecosystem** that makes it "expensive" for a user to leave (accrued value).

*Chart 3: Top Desired Loyalty Benefits*




- **What it shows:** The "No-Fee Monthly Pass" (38%) and "Streak Bonuses" (29%) are the clear winners.
- **The C-Suite Pitch:** To stop the "leaky bucket," we should pivot our marketing spend. Instead of mass-distribution 20% coupons, we should invest in a **subscription model (The No-Fee Pass)** and **gamified retention (Streak Bonuses)**.

*Chart 4: The "Commodity Trap" (Disappointment Metric)*




- **What it shows: 65%** of our "Nomad" customers would **not** be disappointed if our service disappeared tomorrow.
- **The C-Suite Pitch:** This is our biggest risk. We are currently a **utility, not a brand**. We are winning on **price** (which is expensive for us) rather than on **relationship** or **unique value**.

## *Executive Strategy Summary (The "So What?")*

Based on these projected findings, our recommendation to the C-Suite is to consider one of the following:

1. **Shift to Subscription:** Launch a pilot for a "No-Fee Pass." This locks the user into our ecosystem and removes the "per-order friction" that causes them to check other apps.
2. **Gamify Retention:** Implement "Streak Bonuses" (e.g., 3 orders in 14 days = 10% credit). This targets the "Nomad" behavior and rewards consistency rather than one-off discount hunting.
3. **Maximize "Reliability Score":** Focus the Product team on ETA accuracy over raw speed. If we can guarantee an arrival time more accurately than competitors, we become the "safe choice" for high-stakes meals (e.g., family dinners, office lunches).

# A/B Test

Analysis of the survey results suggests that **at least one change** to the way customers interact with the **TableGram** app will improve retention and reduce churn. They also suggest that our current competitive edge is not sustainable, because it revolves around **low prices** that reduce our margins and do not offer a **qualitative benefit** over other products in the same space. Three possible interventions emerged from our analysis, as detailed in the preceding summary; we propose to implement one of them on a trial basis and assess its effect on performance through an **A/B test** with two randomly selected pools of users.

## Proposal

Of the three possible interventions above, we determined that the **gamification** strategy would represent the most **lightweight** option for our developers to implement quickly and with the least impact on the app's overall performance.

Therefore, we tested a new feature that we are labeling "**Streak Bonuses**:" users who complete **3 orders within 14 days** receive **10% back** in app credit to be applied to future orders.

The test was designed as follows:

| Parameter | Value |
|---|---|
| Group A (Control) | **Current** app experience, no streak mechanic. **100 users**. |
| Group B (Variant) | **Streak Bonus enabled**: visible progress tracker, 10% credit after 3 orders in 14 days. **100 users**. |
| Duration | 6 weeks |
| Primary Metric | Order frequency (total orders per user over 6 weeks) |
| Assignment | Random, stratified by baseline order frequency to ensure **balanced** groups. |

## Hypotheses

**Null Hypothesis ($H_0$):** There is **no difference** in mean order frequency between Group A and Group B. Any observed difference is due to chance.

**Alternative Hypothesis ($H_1$):** Group B (Streak Bonus) has a **significantly higher** mean order frequency than Group A.

# A/B Test Quantitative Analysis

## *Statistical Test: Independent Samples t-Test*

We are comparing the means of two independent groups on a continuous outcome (order count). An **independent two-sample *t*-test** is the standard tool for this.

**Formula (Welch's *t*-test):**

$$t = \frac{\Delta \overline{X}}{s_{\Delta \bar{X}}} = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{s^2_{\bar{X}_1} + s^2_{\bar{X}_2}}},$$

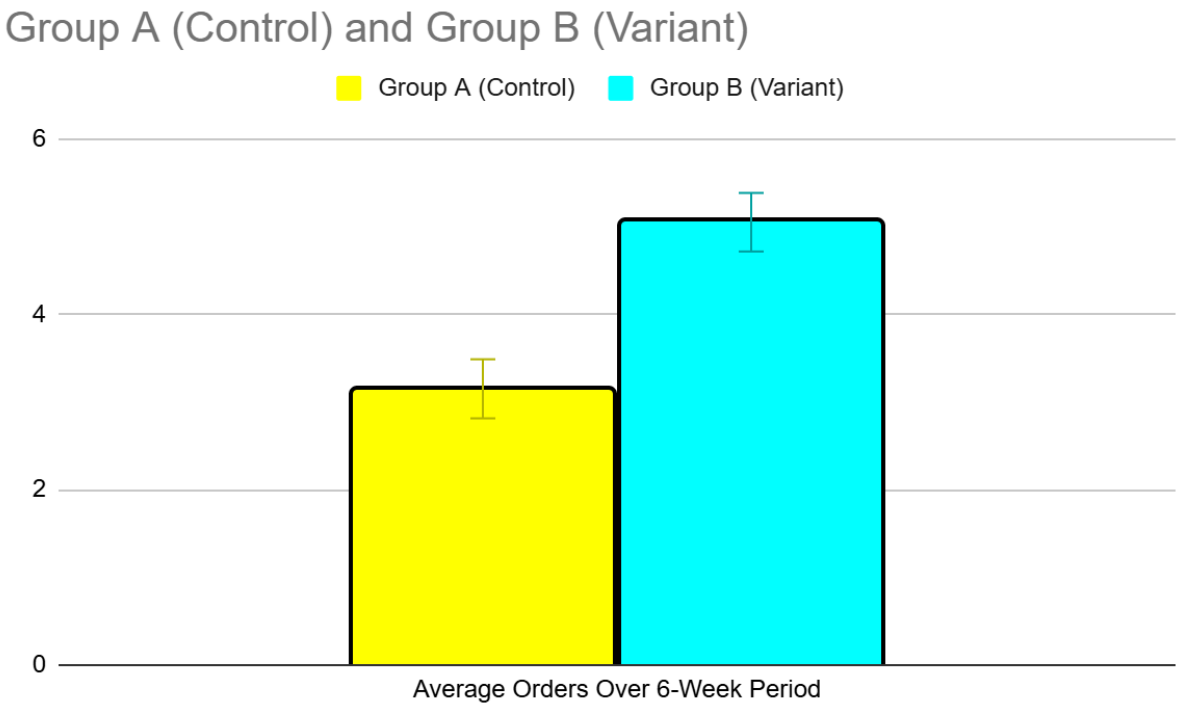
In plain terms: we divide the **difference** between the two group means by a measure of how much **natural variation** exists in the data. If the difference is **large** relative to the **noise**, the ***t*-value** is large and the ***p*-value** is small.

## *Assumptions We Need to Check*

| Assumption | What It Means | What If It Fails? |
|---|---|---|
| Independence | Each user's data is unrelated to other users' data. Guaranteed by random assignment at the account level. | If violated (e.g., shared accounts), the test is invalid. This is a design issue, not a statistical fix. |
| Normality | The order counts in each group should be roughly normally distributed. We check with a Shapiro-Wilk test or by looking at histograms. | With 100 users per group, the Central Limit Theorem makes the *t*-test robust to moderate non-normality. Only severely skewed data is a problem. |
| Equal Variances | Both groups should have similar spread. We check with Levene's test. | If variances are unequal, we use Welch's *t*-test (a slight modification that doesn't assume equal variances). Safe to use Welch's by default. |

## *Results: Data & Visualization (simulated)*

Full simulated **A/B test data + Analysis results** are in a spreadsheet [here]




| Metric | Value |
|---|---|
| Group A Mean (Control) | **3.15** orders over 6 weeks |
| Group B Mean (Streak Bonus) | **5.05** orders over 6 weeks |
| Difference | **1.90** additional orders |
| *p*-value | **< .001** (well below the .05 threshold) |
| 95% Confidence Interval | **[1.56, 2.24]** additional orders |

**Interpretation:** Users with the Streak Bonus placed about **1.9 more** orders over 6 weeks than users without it. This difference is statistically significant ($p < .001$), meaning there is less than a 0.1% chance we would see a gap this large if the Streak Bonus had no real effect. The confidence interval tells us we can be 95% sure the **true effect** for our full user base falls **between 1.56 and 2.24** additional orders.

## *Effect Size: How Big Is This?*

A *p*-value tells us the effect is real, but **doesn't** tell us how *big* the effect is. For that insight, we need to calculate Cohen's *d*:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s}.$$

***d*** = Difference (in means) / Pooled Standard Deviation = 1.90 / 1.71 = **1.11**

Benchmarks for Cohen's *d* analysis:

*d* = 0.2 = small

*d* = 0.5 = medium

*d* = 0.8 = large

Our ***d* = 1.11** is a **large** effect. The Streak Bonus didn't just nudge behavior; it **shifted** ordering patterns by more than one standard deviation. This is a strong practical result that justifies a **full rollout**.

# Executive Summary & Recommendations

Every analysis step exists to answer a **specific business question**. The table below maps the full pipeline from method to decision.

| Analysis | Data Source | What We Learn | Business Decision |
|---|---|---|---|
| Frequency counts & percentages | Survey (all Qs) | What behaviors and preferences are most common | Which loyalty barriers to address first |
| *Chi*-square test | Survey Q1 x Q3, Q3 x Q4 | Whether different customer segments switch for different reasons | Do we need segmented retention strategies or one approach? |
| Rank comparison | Survey Q2 | Which app-selection drivers matter most | Product priority: invest in ETA accuracy, speed, or cost? |
| Proportion comparison | Survey Q5 | Which friction types are most frequent | Top operational fix (e.g., fee transparency vs. delivery speed) |
| Disappointment benchmark | Survey Q7 | How far we are from product-market fit | Urgency level for loyalty investment (C-suite risk metric) |
| Independent *t*-test | A/B test | Whether Streak Bonuses increase order frequency | Go/no-go on full rollout |
| Cohen's *d* (effect size) | A/B test | How large the effect is in practical terms | Scale of investment: aggressive rollout vs. cautious expansion |
| Confidence interval | A/B test | Range of likely true effect for full user base | Revenue forecasting: expect 1.56–2.24 extra orders per user per 6 weeks |

This plan ensures that every number we report connects to a decision. The survey tells us **what to fix** and **for whom**. The A/B test tells us **whether our fix works** and **by how much**.

# Key Takeaways

**Problem**: Low order frequency and lacklustre customer loyalty.

**Solution**: Introduce “Streak Bonuses”

**Impact**: A 60% increase in average orders with high statistical confidence ($P < 0.001$).

By adding the “Streak Bonus” feature to the user experience, we saw a **statistically significant 60% increase** in average order frequency ($p < 0.001$). We ran an A/B test on a pool of 200 users and found that our proposed change resulted in greater customer loyalty, with users moving from an average of **around 3 orders** in a 6-week period to **over 5 orders** and transforming their experience from “utility” to “habit.” This data-driven update to the app effectively addresses the primary business goal of increasing long-term customer loyalty and lifetime value.

# Verdict: Implement Streak Bonus Feature

The results of our A/B test provided conclusive evidence ($p < 0.001$) that users’ average order frequency increases significantly when streak bonuses are added. With a 95% confidence interval of [1.56, 2.24], we can expect approximately **two additional orders per user every six weeks**.

*Recommendation: Transition the variant (Group B) version of the app, with “streak bonus” implemented, from the test environment to a full rollout for 100% of the user base.*

# Final Considerations

While the data suggest that the “streak bonus” will improve our customer loyalty, it is essential that we **continuously monitor** the app’s performance and consider **other factors** that we may not be able to understand fully with the testing data available. Furthermore, we must also be cognizant of the possibility that **other changes** to the user experience for TableGram could produce **additional improvements** in customer loyalty and total revenue.

Therefore, we conclude with the following considerations regarding the future of the TableGram product:

## *Insight: Loyalty Threshold*

The data show that Group B users went from making an average of ~3 orders to ~5 orders. In the food delivery space, **5+ orders** can be seen as a psychological threshold beyond which the app becomes a **habit** rather than just a **utility**.

*Action: Analyze the specific pieces of the new design that led to the 4th and 5th orders. How accessible/visible is the "streak bonus feature" for new vs. returning users? How easy is it to monitor their progress? Was every member of Group B aware of it? If any were not, why was that the case?*

*Application: Implement these specific UI/UX elements in other areas of the app.*

## *Variance Question*

While Group B users demonstrated greater loyalty to the app, with an average of just under 2 additional orders over the 6-week trial period, the **standard deviation in this group was effectively the same** as with the control group (both around 1.7). This variance suggests that the streak bonus feature was a hit with some but not all users; perhaps it was confusing, unexpected, or not even noticed by this subset.

*Action: Conduct a follow-up qualitative focus group with users who fell on the lower end of Group B (those who ordered 3 or fewer times).*

*Application: Refine the UI to reduce the cognitive load for the lower-frequency users in order to ensure that streak bonuses are accessible for everyone, not just regulars.*

## *Gaming the System*

Some users may place **cheap minimum orders** just to hit the streak, producing little to no improvement in total order revenues as a result of the new feature. It's even possible that users order **less value** in food overall because their focus in using the app has shifted from its **utility** as a food delivery to the **gamified usage metrics** and the desire to "win" discounts.

*Action: Check whether the average order value dropped in Group B compared to Group A.*

## *Novelty Effect*

While our data indicate that introducing streak bonuses to Group B produced greater customer loyalty to the app, it is possible that this increase in usage was a function of the feature's novelty; in other words, some users may have preferred the new feature **simply because it was new.**

*Action: Schedule a longitudinal study 3-6 months after rollout to determine whether the increased loyalty remains stable after streak bonuses are no longer new.*

*Metric: Monitor the churn rate of Group B users compared to the pre-rollout average.*

## *Future Directions*

Given that streak bonuses appear to have been successful at driving volume, we see opportunities to implement **additional** features or changes in the future.

*Personalization: User rewards could be tailored to individual users' tastes; for instance, a demonstrated preference for Thai could earn a special discount applicable to Thai food after a certain number of orders.*

*Variety: As an alternative, user rewards could also incentivize trying new and different cuisines; for example, users could "collect" different cuisines and receive discounts for their adventurous palate.*